

LA HW

Included exercises: 1, 3, 9, 11, 17, 21, 25, 27, 31, 33, 35, 37, 39, 41, 45, 49, 53, 57, 61, 63, 69, 71

Exercises 1–8: Membership in a Span

Determine whether the given vector belongs to

$$\text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \right\}.$$

Original Exercise 1 $\begin{bmatrix} -1 \\ 4 \\ 7 \end{bmatrix}.$

Original Exercise 3 $\begin{bmatrix} 0 \\ 5 \\ 2 \end{bmatrix}.$

Exercises 9–16: Membership in a Span

Determine whether the given vector belongs to

$$\text{Span} \left\{ \begin{bmatrix} 1 \\ 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 0 \\ 3 \end{bmatrix} \right\}.$$

Original Exercise 9 $\begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \end{bmatrix}.$

Original Exercise 11 $\begin{bmatrix} -8 \\ 9 \\ -7 \\ 2 \end{bmatrix}.$

Exercises 17–20: Parameterized MembershipDetermine the values of r for which $\mathbf{v}$ belongs to the span of S .**Original Exercise 17**

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix} \right\}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ r \\ -1 \end{bmatrix}.$$

Exercises 21–28: Generating Sets

Determine whether each set is a generating set for the indicated space $\mathbb{R}^n$.

Original Exercise 21

$$\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \end{bmatrix} \right\} \subset \mathbb{R}^2.$$

Original Exercise 25

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} \right\} \subset \mathbb{R}^3.$$

Original Exercise 27

$$\left\{ \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ -7 \\ 2 \end{bmatrix}, \begin{bmatrix} -5 \\ 7 \\ 6 \end{bmatrix} \right\} \subset \mathbb{R}^3.$$

Exercises 29–36: Consistency for Every Right-Hand Side

For each $m \times n$ matrix A , determine whether $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^m$.

Original Exercise 31 $A = \begin{bmatrix} 1 & 0 & -3 \\ -1 & 0 & 3 \end{bmatrix}.$

Original Exercise 33 $A = \begin{bmatrix} 1 & -1 \\ 0 & 1 \\ -2 & 2 \end{bmatrix}.$

Original Exercise 35 $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 6 \end{bmatrix}.$

Exercises 37–44: Smallest Subset with the Same Span

For each set S , find a subset of S having the same span as S and containing as few vectors as possible.

Original Exercise 37

$$S = \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}.$$

Original Exercise 39

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Original Exercise 41

$$S = \left\{ \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 4 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

Exercises 45–64: True or False

Determine whether each statement is true or false.

Original Exercise 45 Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$ be a nonempty set of vectors in $\mathbb{R}^n$. A vector $\mathbf{v}$ belongs to $\text{Span } S$ if and only if $\mathbf{v} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k$ for some scalars $c_1, c_2, \dots, c_k$.

Original Exercise 49 Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\} \subset \mathbb{R}^n$. Then $\text{Span } S = \mathbb{R}^n$ if and only if $\text{rank}[\mathbf{u}_1 \ \mathbf{u}_2 \ \dots \ \mathbf{u}_k] = n$.

Original Exercise 53 If finite subsets $S_1, S_2 \subset \mathbb{R}^n$ have equal spans, then S_1 and S_2 contain the same number of vectors.

Original Exercise 57 If $\mathbf{v} \in \text{Span } S$, then $c\mathbf{v} \in \text{Span } S$ for every scalar c .

Original Exercise 61 If an $m \times n$ matrix A has a reduced row echelon form containing no zero rows, then the columns of A form a generating set for $\mathbb{R}^m$.

Original Exercise 63 If A is an $m \times n$ matrix such that $A\mathbf{x} = \mathbf{b}$ is inconsistent for some $\mathbf{b} \in \mathbb{R}^m$, then $\text{rank } A < m$.

Exercises 69 and 71: Proofs About Span

Supply the requested proofs.

Original Exercise 69 Let S_1 and S_2 be finite subsets of $\mathbb{R}^n$ with $S_1 \subseteq S_2$. Prove that if S_1 is a generating set for $\mathbb{R}^n$, then S_2 is also a generating set for $\mathbb{R}^n$.

Original Exercise 71 Let $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k \in \mathbb{R}^n$ and let $c_1, c_2, \dots, c_k$ be nonzero scalars. Prove that

$$\text{Span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\} = \text{Span}\{c_1\mathbf{u}_1, \dots, c_k\mathbf{u}_k\}.$$